

Feedback — Module 1

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You submitted this quiz on **Sun 12 Jul 2015 11:26 AM PDT**. You got a score of **10.00** out of **10.00**.

Question 1

Which of the following is not a suffix of CATATTAC?

Your Answer	Score	Explanation
☐ C		
☐ TAC		
☐ TATTAC		
☒ CAT	✓ 1.00	
Total	1.00 / 1.00	

Question 2

What's the longest prefix of CACACTGCACAC that is also a suffix?

Your Answer	Score	Explanation
☐ CACACTG		
☒ CACAC	✓ 1.00	
☐ CAC		
☐ C		
Total	1.00 / 1.00	

Question 3

Which of the following is not a substring of GCTCAGCGGGGCA?

Your Answer	Score	Explanation
☐ GCG		
☐ GCA		
☐ GCT		
☒ GCC	✓ 1.00	
Total	1.00 / 1.00	

Question 4

Starting around 2007, the cost of DNA sequencing started to decrease rapidly because more laboratories started to use:

Your Answer	Score	Explanation
☐ Sanger sequencing		
☐ DNA microarrays		
☒ Second-generation sequencing	✓ 1.00	
☐ Double sequencing		
Total	1.00 / 1.00	

Question 5

Which of the following pieces of information is not included in a sequencing read in the FASTQ format:

Your Answer	Score	Explanation
☒ Which chromosome the read originated from	✓ 1.00	

- ☐ A "name" for the read
- ☐ The sequence of base qualities corresponding to the bases
- ☐ The sequence of bases that make up the read

Total	1.00 / 1.00
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Question 6

If read alignment is like "looking for a needle in a haystack," then the "haystack" is the:

Your Answer	Score	Explanation
☒ Reference genome	✓ 1.00	
☐ Sequencing read		
☐ Sequencer		
☐ Gene database		
Total	1.00 / 1.00	

Question 7

The Human Genome Project built the initial "draft" sequence of the human genome, starting from sequencing reads. The computational problem they had to solve was the:

Your Answer	Score	Explanation
☐ read alignment problem		
☐ prime factorization problem		
☒ de novo shotgun assembly problem	✓ 1.00	
☐ gene finding problem		
Total	1.00 / 1.00	

Question 8

If the length of the pattern is x and the length of the text is y , the minimum possible number of character comparisons performed by the naive exact matching algorithm is:

Your Answer	Score	Explanation
☐ xy		
☒ $y - x + 1$	✓ 1.00	
☐ $x(y - x + 1)$		
☐ $x + y$		
Total	1.00 / 1.00	

Question 9

If the length of the pattern is x and the length of the text is y , the maximum possible number of character comparisons performed by the naive exact matching algorithm is:

Your Answer	Score	Explanation
☐ $y - x + 1$		
☐ xy		
☐ $x + y$		
☒ $x(y - x + 1)$	✓ 1.00	
Total	1.00 / 1.00	

Question 10

Say we have a function that generates a random DNA string, i.e. the kind of string we would get by rolling a 4-sided die (A/C/G/T) over and over. We use the function to generate a random pattern P of length 20 and a random text T of length 100. Now we run the naive exact matching algorithm to find matches of P within T . We expect the total number of character comparisons we

perform to be closer to the...

Your Answer	Score	Explanation
☒ minimum possible	✓ 1.00	
☐ maximum possible		
Total	1.00 / 1.00	